

Package Insert

1. Product name

Gabenil 300 Capsules

Gabenil 400 Capsules

2. Name and Strength of the Active Ingredient

Each Gabenil 300 mg capsule contains 300 mg Gabapentin.

Each Gabenil 400 mg capsule contains 400 mg Gabapentin.

3. Product description

Gabenil 300 are yellow/yellow capsules.

Gabenil 400 are orange/orange capsules.

4. Pharmacological Properties

The precise mechanism of action of gabapentin is not known.

Gabapentin is structurally related to the neurotransmitter GABA (gamma-aminobutyric acid) but its mechanism of action is different from that of several other active substances that interact with GABA synapses, including valproate, barbiturates, benzodiazepines, GABA transaminase inhibitors, GABA uptake inhibitors, GABA agonists, and GABA prodrugs. *In vitro* studies with radiolabeled gabapentin have characterized a novel peptide binding site in rat brain tissues including neocortex and hippocampus that may relate to anticonvulsant and analgesic activity of gabapentin and its structural derivatives.

The binding site for gabapentin has been identified as the α_2 -delta subunit of voltage-gated calcium channels.

Pharmacodynamics

Pharmacotherapeutic group: Antiepileptics, ATC Code: N03AX12

Gabapentin at relevant clinical concentrations does not bind to other common drug or neurotransmitter receptors of the brain including GABA_A, GABA_B, benzodiazepine, glutamate, glycine or N-methyl-d-aspartate receptors.

Gabapentin does not interact with sodium channels *in vitro* and so differs from phenytoin and carbamazepine. Gabapentin partially reduces responses to the glutamate agonist N-methyl-D-

aspartate (NMDA) in some test systems *in vitro*, but only at concentrations greater than 100 μM , which are not achieved *in vivo*. Gabapentin administration to rats increases GABA turnover in several brain regions in a manner similar to valproate sodium, although in different regions of the brain. The relevance of these various actions of gabapentin readily enters the brain and prevents seizures from maximal electroshock, from chemical convulsants including inhibitors of GABA synthesis, and in genetic models of seizures.

Pharmacokinetics

Absorption

Following oral administration, peak plasma gabapentin concentrations are observed within 2 to 3 hours. Gabapentin's bioavailability (fraction of dose absorbed) tends to decrease with increasing dose. Absolute bioavailability of a 300 mg capsule is approximately 60%. Food, including a high-fat diet, has no clinical significant effect on gabapentin pharmacokinetics.

Gabapentin pharmacokinetics are not affected by repeated administration. Although plasma gabapentin concentrations were generally between 2 $\mu\text{g/ml}$ and 20 $\mu\text{g/ml}$ in clinical studies, such concentrations were not predictive of safety or efficacy.

Pharmacokinetic parameters are given in Table 1.

Table 1						
Pharmacokinetic Parameter	300 mg (N=7)		400 mg (N=14)		800 mg (N=14)	
	Mean	%CV	Mean	%CV	Mean	%CV
C _{max} ($\mu\text{g/ml}$)	4.02	(24)	5.74	(38)	8.71	(29)
T _{max} (hr)	2.7	(18)	2.1	(54)	1.6	(76)
T _{1/2} (hr)	5.2	(12)	10.8	(89)	10.6	(41)
AUC (0-8) $\mu\text{g} \cdot \text{hr/ml}$	24.8	(24)	34.5	(34)	51.4	(27)
Ae% (%)	NA	NA	47.2	(25)	34.4	(37)

C_{max}= Maximum steady state plasma concentration

T_{max}= Time for C_{max}

T_{1/2}= Elimination half-life

AUC (0-8) = Steady state area under plasma concentration-time curve from time 0 to 8 hours post dose

Ae% Percent of dose excreted unchanged into the urine from time 0 to 8 hours post dose

NA = Not available

Distribution

Gabapentin is not bound to plasma protein and has a volume of distribution equal to 57.7 liters. In patients with epilepsy, gabapentin concentrations in cerebrospinal fluid (CSF) are approximately 20% of corresponding steady-state through plasma concentrations. Gabapentin

is present in the breast milk of breast-feeding women.

Metabolism

There is no evidence of gabapentin metabolism in humans. Gabapentin does not include hepatic mixed function oxidase enzymes, responsible for drug metabolism.

Elimination

Gabapentin is eliminated unchanged solely by renal function. The elimination half-life is independent of dose and averages of 5 to 7 hours.

In elderly patients, and in patients with impaired renal function, gabapentin plasma clearance is reduced. Gabapentin elimination-rate is constant, plasma clearance and renal clearance are directly proportional to creatinine clearance.

Gabapentin is removed from plasma by hemodialysis. Dosage adjustment in patients with compromised renal function or undergoing hemodialysis is recommended.

Gabapentin pharmacokinetics in children were determined in 50 healthy subjects between the ages of 1 month and 12 years. In general, plasma gabapentin concentrations in children above five years of age are similar to those in adults, when dosed on a mg/kg basis.

Linearity/ Non linearity

Gabapentin bioavailability (fraction of dose absorbed) decreases with increasing dose with imparts non-linearity to pharmacokinetic parameters which do not include F, such as CL_r and T_{1/2} are best described by linear pharmacokinetics. Steady state plasma gabapentin concentrations are predictable from single dose data.

5. Clinical Particulars

Indications

Epilepsy

Gabapentin is indicated as adjunctive therapy in the treatment of partial seizures with and without secondary generalization in adults and children aged 6 years and above.

Gabapentin is indicated as monotherapy in the treatment of partial seizures with and without secondary generalization in adults and adolescents aged 12 years and above.

Treatment of peripheral neuropathic pain

Gabapentin is indicated for the treatment of peripheral neuropathic pain, which includes diabetic neuropathy, post-herpetic neuralgia and trigeminal neuralgia in adults age 18 years and above. Safety and effectiveness in patients below the age of 18 years have not been established.

Posology and method of administration

For all indications a titration scheme for the initiation of therapy is described in Table 2, which is recommended for adults and adolescents aged 12 years and above. Dosing instructions for children under 12 years of age are provided under a separate sub-heading later in this section.

Table 2			
DOSAGE CHART – INITIAL TITRATION			
Dose	Day 1	Day 2	Day 3
900 mg	300 mg QD ^a	300 mg BID ^b	300 mg TID ^c

QD^a = once a day

BID^b = two times a day

TID^c = three times a day

Discontinuation of gabapentin:

In accordance with current clinical practice, if gabapentin has to be discontinued it is recommended this should be done gradually over a minimum of one week independent of the indication.

Epilepsy:

Epilepsy typically requires long-term therapy. Dosage is determined by the treating physician according to individual tolerance and efficacy.

Adults and adolescents:

In clinical trials, the effective dosing range was 900 to 3600 mg/day. Therapy may be initiated by titrating the dose as described in table 1 or by administering 300 mg three times a day (TID) on Day 1. Thereafter, based on individual patient response and tolerability, the dose can be further increased in 300 mg/day increments every 2-3 days up to maximum dose of 3600 mg/day. Slower titration of gabapentin dosage may be appropriate for individual patients. The minimum time to reach a dose of 1800 mg/day is one week, to reach 2400 mg/day is a total of 2 weeks and to reach 3600 mg/day is a total of 3 weeks. Dosages up to 4800 mg/day have been well tolerated in long-term open-label clinical studies. The total daily dose should be divided in three single doses, the maximum time interval between the doses should not exceed 12 hours to prevent breakthrough convulsions.

Children aged 6 years and above:

The starting dose should range from 10 to 15 mg/kg/day and the effective dose is reached by upward titration over a period of approximately three days. The effective dose of gabapentin in children aged 6 years and older is 25 to 35 mg/kg/day. Dosages up to 50 mg/kg/day have been well tolerated in a long-term clinical study. The total daily dose should be divided in three single doses, the maximum time interval between doses should not exceed 12 hours.

It is not necessary to monitor gabapentin plasma concentrations to optimize gabapentin therapy. Further, gabapentin may be used in combination with other antiepileptic medicinal products without concern for alteration of the plasma concentrations of gabapentin or serum concentrations of other antiepileptic medicinal products.

Peripheral Neuropathic Pain:

Adults:

The therapy may be initiated by titrating the dose as described in Table 2. Alternatively, the starting dose is 900 mg/day given as three equally divided doses. Thereafter, based on individual patient response and tolerability, the dose can be further increased in 300 mg/day increments every 2-3 days up to a maximum dose of 3600 mg/day.

Elderly (over 65 years of age):

Elderly patients may require dosage adjustment because of declining renal function with age (see Table 2). Somnolence, peripheral oedema and asthenia may be more frequent in elderly patients.

Renal impairment:

Dosage adjustment is recommended in patients with compromised renal function as described in Table 3 and/or those undergoing haemodialysis

Table 3	
DOSAGE OF GABAPENTIN IN ADULTS BASED ON RENAL FUNCTION	
Creatinine Clearance (mL/min)	Total Daily Dose ^a (mg/day)
≥80	900-3600
50-79	600-1800
30-49	300-900
15-29	150 ^b -600
<15	150 ^b -300

^a Total daily dose should be administered as three divided doses. Reduced dosages are for patients with renal impairment (creatinine clearance < 79 mL/min).

^b The 150 mg daily dose to be administered as 300 mg every other day.

Use in patients undergoing haemodialysis:

For anuric patients undergoing hemodialysis who have never received gabapentin, a loading dose of 300 to 400 mg, then 300 mg of gabapentin following each 4 hours of hemodialysis, is recommended.

Method of administration:

For oral use.

Gabapentin should be given with or without food and should be swallowed whole with sufficient fluid-intake (e.g. a glass of water).

When to use it

Gabenil is usually given in 3 separate doses, by taking the capsule(s) each day, usually once in the morning, once in the afternoon and once in the evening.

How long to use it

Continue taking Gabenil until your doctor tells you to stop.

If you forget to use it

If you forget to take a dose, take it as soon as you remember unless it is time for your next dose. Do not take a double dose to make up for a forgotten dose.

Contraindications

When you must not use it

The use of this medicine is contraindicated to patients with known hypersensitivity to gabapentin or to any of the excipients of the product.

Before you start use it

Special caution is recommended for patients with a history of psychotic episodes.

Patients taking gabapentin can be the subject of mood and behavioural disturbances. Potential for an increase in risk of suicidal thoughts and behaviours.

Gabril is not generally considered effective in the treatment of absence seizures.

Important information about some of the ingredients of Gabril

Gabril capsules contain lactose (a type of sugar). If you have been told by your doctor that you have an intolerance to some sugars, contact your doctor before taking this medicinal product.

Things you must not do

Driving and using machines

Gabapentin may cause drowsiness, dizziness or other related symptoms. You should not drive, operate complex machinery or take part in other potentially hazardous activities until you know whether this medication affects your ability to perform these activities.

Warnings and Precautions

Things to be careful of

If a patient develops acute pancreatitis under treatment with gabapentin, discontinuation of gabapentin should be considered.

Although there is no evidence of rebound seizures with gabapentin, abrupt withdrawal of anticonvulsants in epileptic patients may precipitate status epilepticus. Any dose reduction, discontinuation or substitution of alternative anticonvulsant medication should be effected gradually over a period of at least one week.

Patients who require concomitant treatment with morphine may experience increases in gabapentin concentrations. Patients should be carefully observed for signs of CNS depression, such as somnolence and the dose of gabapentin or morphine should be reduced appropriately.

Interactions with other medicaments

Please tell your doctor or pharmacist if you are taking or have recently taken any other medicines, including medicines obtained without a prescription.

Co-administration of gabapentin with oral contraceptives containing norethindrone and/or ethinyl estradiol, does not influence the steady-state pharmacokinetics of either component.

Renal excretion of gabapentin is unaltered by probenecid.

A slight decrease in renal excretion of gabapentin that is observed when it is co-administered with cimetidine is not expected to be of clinical importance.

Concomitant administration with morphine increases the mean gabapentin AUC. Therefore, patients should be carefully observed for signs of CNS depression, such as somnolence, and the dose of gabapentin or morphine should be reduced appropriately.

Co-administration of gabapentin with antacids containing aluminium and magnesium reduces gabapentin bioavailability up to 24%. It is recommended that gabapentin be taken at the earliest two hours following antacid administration.

No interaction between gabapentin and phenobarbital, phenytoin, valproic acid, or carbamazepine has been observed.

Gabapentin steady-state pharmacokinetics are similar for healthy subjects and patients with epilepsy receiving these antiepileptic agents.

Pregnancy and Lactation

Safe use in humans has not been established. Therefore, Gabenil should be avoided during pregnancy.

Gabapentin is excreted into human milk and therefore it is not recommended during lactation. Ask your doctor or pharmacist for advice before taking any medicine.

Undesirable effects

Like all medicines, Gabenil can cause side effects, although not everybody gets them.

The adverse reactions observed during clinical studies conducted in epilepsy (adjunctive and monotherapy) and neuropathic pain have been provided in a single list below by class and frequency (very common ($\geq 1/10$); common ($\geq 1/100$ to $<1/10$); uncommon ($\geq 1/1000$ to $<1/100$); rare ($\geq 1/10000$ to $<1/100$); very rare ($<1/10000$). Where an adverse reaction was seen at different frequencies in clinical studies, it was assigned to the highest frequency reported.

Additional reactions reported from post-marketing experience are included as frequency 'Not known' (cannot be estimated from the available data) in *italics* below.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

System organ class	Adverse drug reactions
Infections and infestations	
Very common	viral infection
Common	pneumonia, respiratory infection, urinary tract infection, infection, otitis media
Blood and the lymphatic system disorders	
Common	leucopenia
Not known	thrombocytopenia
Immune system disorders	
Uncommon	allergic reactions (e.g. urticaria)
Metabolism and nutrition disorders	
Common	anorexia, increased appetite
Psychiatric disorders	
Common	hostility, confusion and emotional lability, depression, anxiety, nervousness, thinking abnormal
Nervous system disorders	
Very Common	somnolence, dizziness, ataxia
Common	convulsions, hyperkinesia, dysarthria, amnesia, tremor, insomnia, headache, sensations such as paresthesia, hypaesthesia, coordination abnormal, nystagmus, increased, decreased, or absent reflexes
Uncommon	hypokinesia
Not known	other movement disorders (e.g. choreoathetosis, dyskinesia, dystonia)
Eye disorders	
Common	visual disturbances such as amblyopia, diplopia
Ear and labyrinth disorders	
Common	vertigo
Not known	tinnitus
Cardiac disorders	
Uncommon	palpitations
Vascular disorders	
Common	hypertension, vasodilatation
Respiratory, thoracic and mediastinal disorders	
Common	dyspnea, bronchitis, pharyngitis, cough, rhinitis
Gastrointestinal disorders	
Common	vomiting, nausea, dental abnormalities, gingivitis, diarrhea, abdominal pain, dyspepsia, constipation, dry mouth or throat, flatulence
Rare	dysphagia
Not known	pancreatitis

Hepatobiliary disorders	
Not known	hepatitis, jaundice
Skin and subcutaneous tissue disorders	
Common	facial oedema, purpura most often described as bruises resulting from physical trauma, rash, pruritus, acne
Not known	Stevens-Johnson syndrome, angioedema, erythema multiforme, alopecia
Musculoskeletal and connective tissue disorders	
Common	arthralgia, myalgia, back pain, twitching
Not known	myoclonus
Renal and urinary disorders	
Not known	acute renal failure, incontinence
Reproductive system and breast disorders	
Common	impotence
Not known	breast hypertrophy, gynaecomastia
General disorders and administration site conditions	
Very common	fatigue, fever
Common	peripheral oedema, abnormal gait, asthenia, pain, malaise, flu syndrome

Under treatment with gabapentin cases of acute pancreatitis were reported. Causality with gabapentin is unclear.

In patients with haemodialysis due to end-stage renal failure, myopathy with elevated creatine kinase levels has been reported.

Respiratory tract infections, otitis media, convulsions and bronchitis were reported only in clinical studies in children. Additionally, in clinical studies in children, aggressive behavior and hyperkinesia were reported commonly.

Reporting of side effects

If you get any side effects, talk to your doctor or pharmacist. This includes any possible side effects not listed in this leaflet.

You may report any side effects or adverse drug reactions directly to the National Centre for Adverse Drug Reaction Monitoring by calling Tel: 03-78835550, or visiting the website portal.bpfk.gov.my (Consumers → Reporting).

Symptoms and treatment of overdose

Acute, life-threatening toxicity has not been observed with gabapentin overdoses of up to 49 grams. Symptoms of the overdoses included dizziness, double vision, slurred speech, drowsiness, loss of consciousness, lethargy and mild diarrhoea. All patients recovered fully with supportive care. Reduced absorption of gabapentin at higher doses may limit drug absorption at the time of overdosing and, hence, minimize toxicity from overdoses.

Although gabapentin can be removed by haemodialysis, it is usually not required. However, in patients with severe renal impairment, haemodialysis may be indicated.

6. Pharmaceutical Particulars

Storage conditions

Keep this medicine out of the sight and reach of children.

Do not use this medicine after the expiry date which is stated on the carton label and blister foil. The expiry date refers to the last day of that month.

Store below 30°C. Protect from light and moisture.

Shelf-life

Gabenil 300 mg capsules

3 years

Gabenil 400 mg capsules

2 years

Pack size available

Gabenil 300 mg capsules

PVC/Aluminium Blisters. Pack sizes of 50, 100 and 1000 capsules.

Gabenil 400 mg capsules

PVC/Aluminium Blisters. Pack size of 50 capsules.

Not all pack sizes may be marketed

7. Manufacturer

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8. Date of revision of the text

09/2020

For internal use: my-pi-gabenil-300mg-400mg-caps-a7